Module 8, Lesson 3

AI 101 for Mechanical Engineers: Imbalance and deployment consequences

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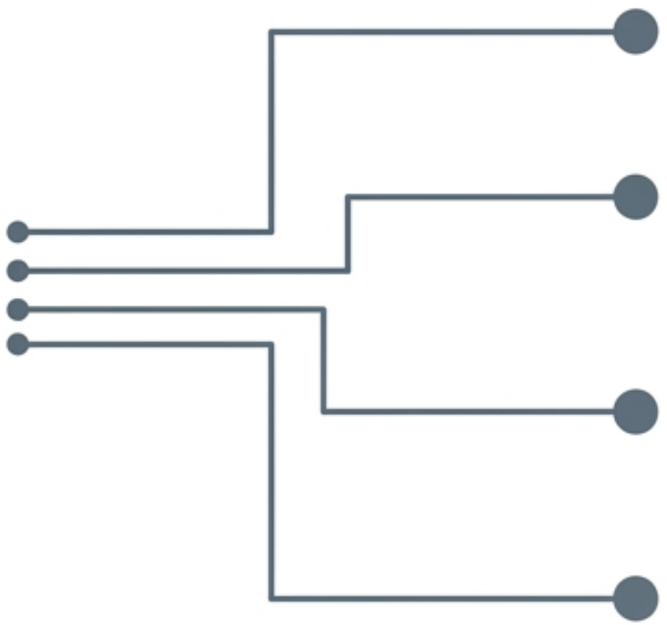
Presentation Structure

1. Imbalance Metrics
2. Threshold Calibration
3. Deployment Workflow
4. Evaluation Standards

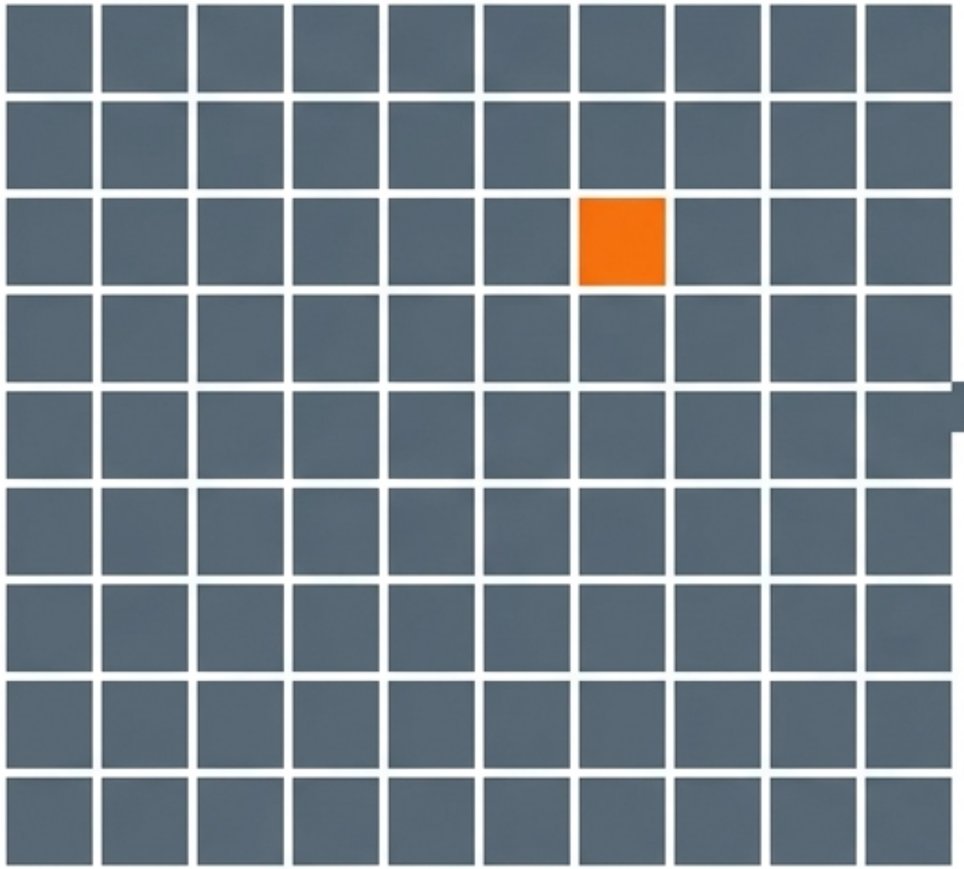
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1. Imbalance Metrics

Quantifying the failure of standard accuracy in skewed datasets.



High accuracy masks rare fault detection



[PREDICT: ALL HEALTHY]

Accuracy: 99%

Fault Recall: 0%

Class imbalance features unequal target prevalence.

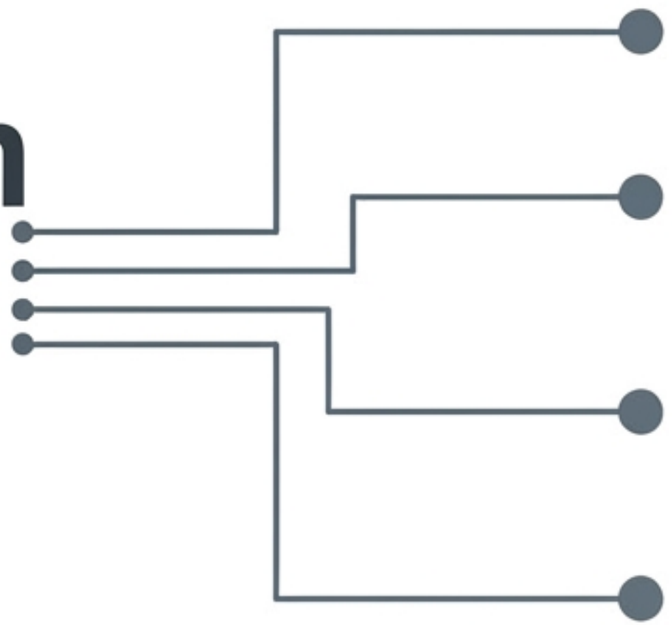
Machinery faults are rare versus healthy periods.

Predicting only the majority class inflates accuracy.

Detectors must outperform the majority baseline.

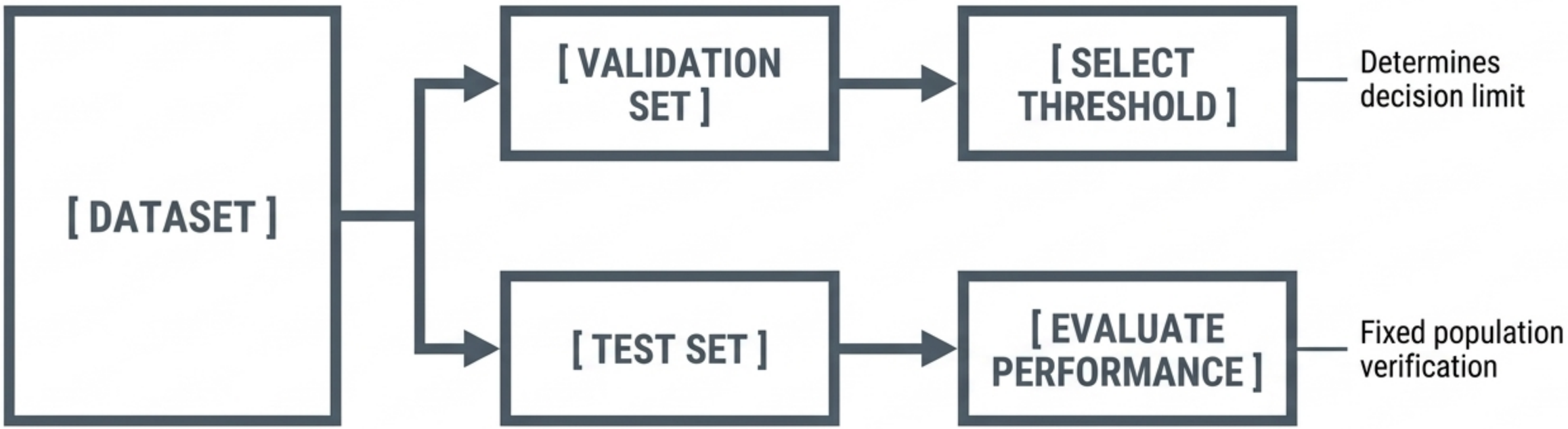
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2. Threshold Calibration



Establishing detection limits using partitioned validation data.

Validation sets define operational thresholds

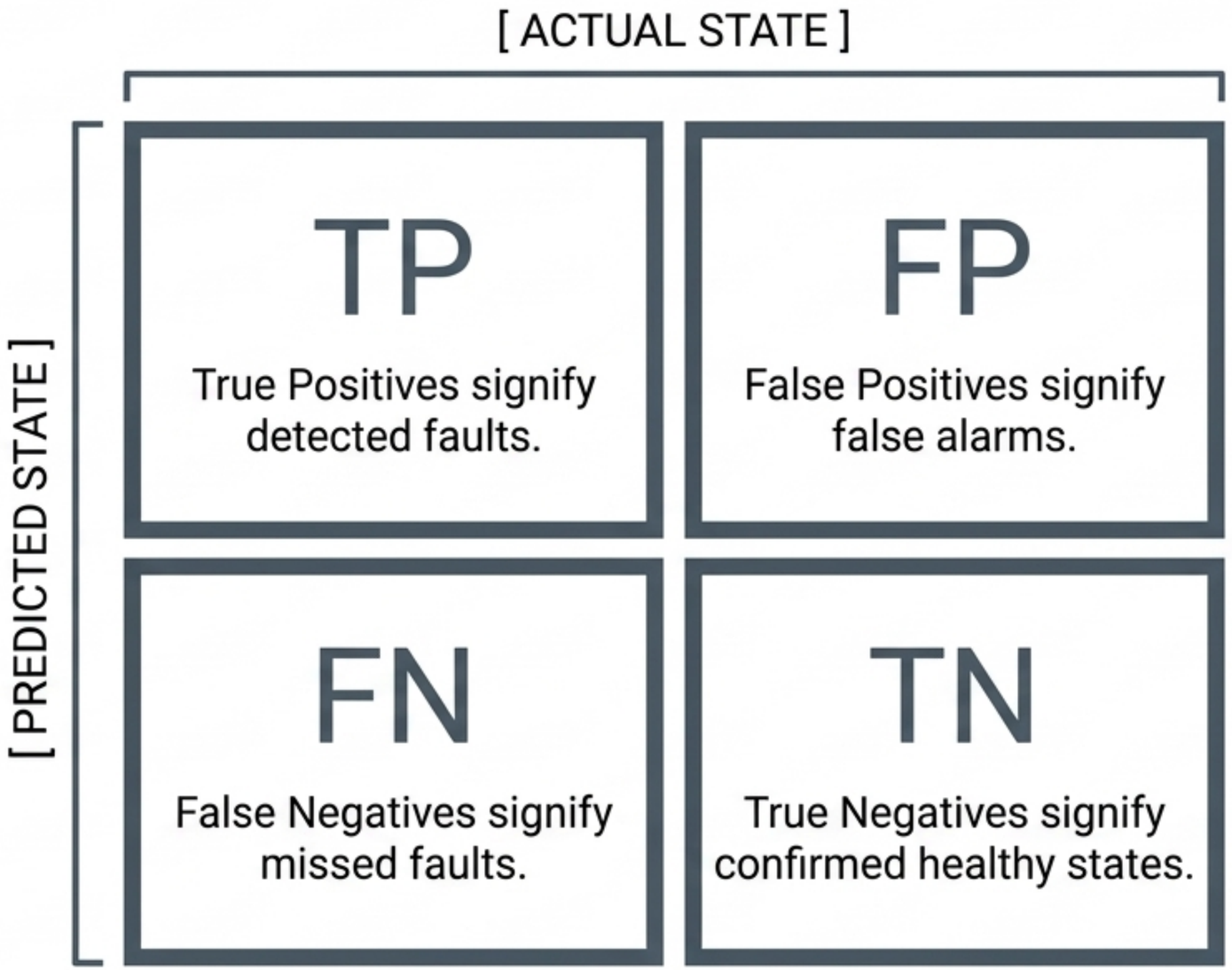


Lab exercises utilize synthetic evaluation scores. Synthetic data cannot validate physical equipment.

Validation examples determine the decision threshold. Fixed test populations verify final system performance.

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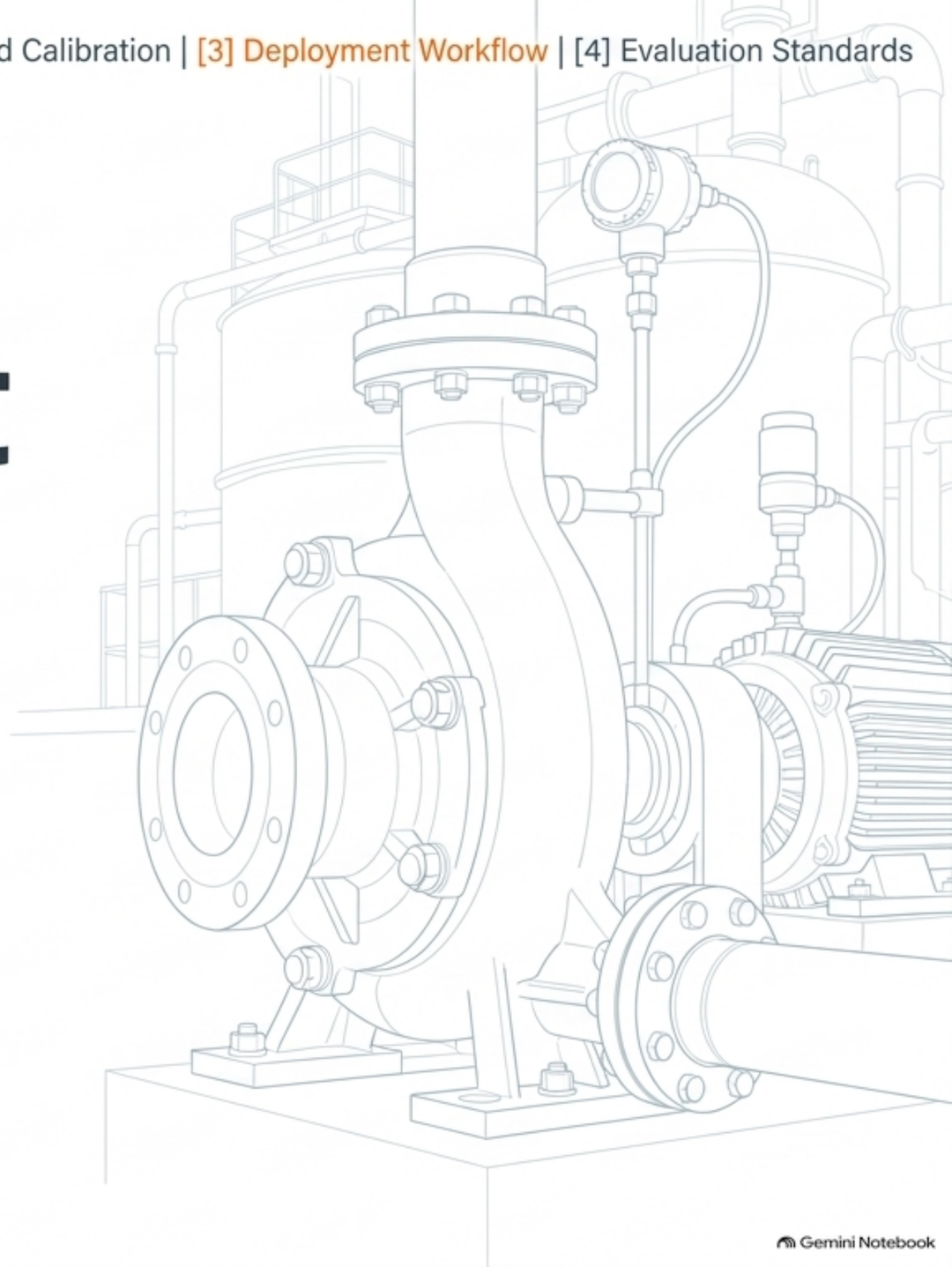
Record all primary prediction outcomes



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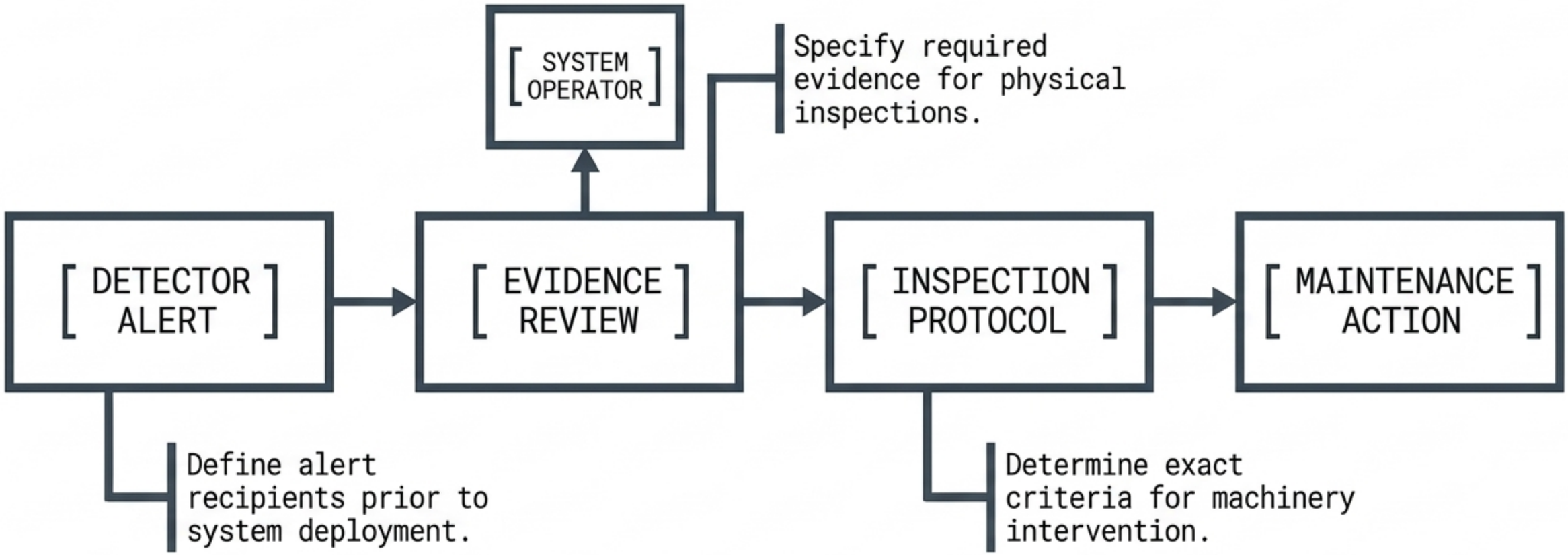
3. Deployment Workflow

Translating algorithmic alerts into mechanical engineering actions.



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Alerts require predefined operational protocols

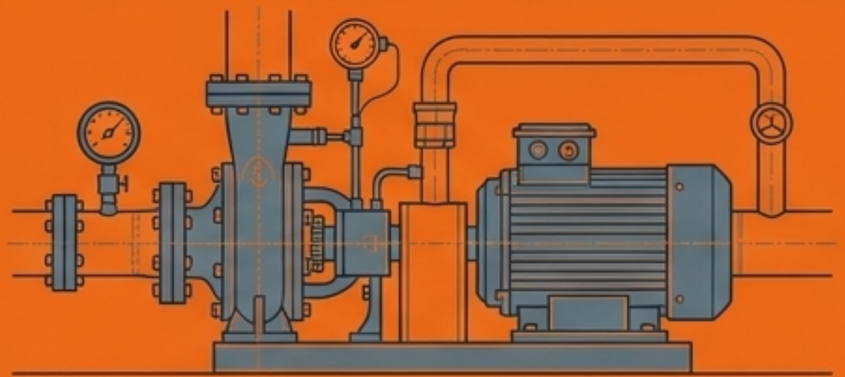


Automatic deployment approval is strictly prohibited.

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Error costs dictate threshold adjustments

[MISSED ALARM (FN)] → [CATASTROPHIC FAILURE]



[FALSE ALARM (FP)] → [RESOURCE DEPLETION]

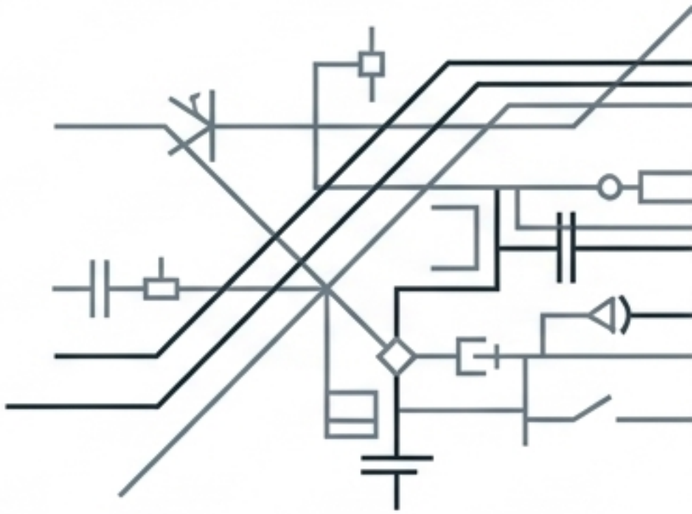
Distinct machinery dictates unique failure costs.

Missed alarms result in catastrophic mechanical failure.

False alarms consume finite maintenance resources.

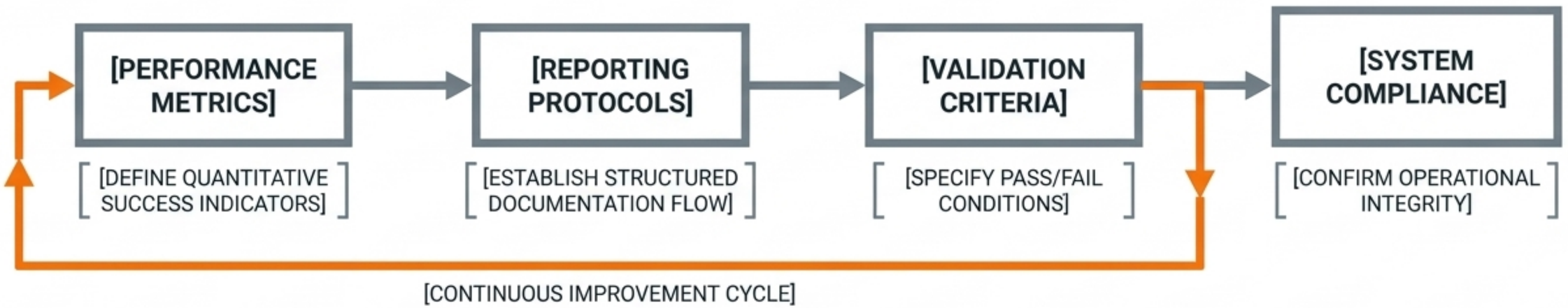
Classroom exercises illustrate these economic trade-offs.

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4. Evaluation Standards

Standardizing terminology and reporting requirements for system validation.



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Precision and recall quantify operational success

$$\text{Precision} = \frac{TP}{TP + FP}$$

[Fraction of alerts that are actual faults]

Measures the accuracy of generated alerts.

$$\text{Recall} = \frac{TP}{TP + FN}$$

[Fraction of actual faults detected]

Measures the fraction of detected faults.

Calculated ratios remain dimensionless quantities. Metrics utilize established classification equations.

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Aggregate accuracy requires supplementary context

Do not report aggregate accuracy alone.



- **[CLASS COUNTS]** — Document exact target class prevalence counts.
- **[MISSED FAULTS (FN)]** — Report all missed faults.
- **[FALSE ALARMS (FP)]** — Report all false alarms.
- **[SELECTED THRESHOLD]** — Specify the exact mathematical threshold selected.

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Key Points

- Imbalance Metrics: Majority-class baselines expose misleading high-accuracy figures.
- Threshold Calibration: Validation data selects thresholds; test populations confirm performance.
- Deployment Workflow: Detector alerts require strict human review and evidence protocols.
- Evaluation Standards: Reporting demands precision, recall, and exact threshold values.

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